MINCHAN KIM

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EDUCATION

Qingdao Daewon School, Qingdao, China Expected June 2027

- Unweighted GPA 3.984 / 4.0. AP exams planned for May 2027: Calculus BC, Physics C (Mechanics; Electricity and Magnetism), Physics 1, Computer Science A.

HONORS AND AWARDS

- 1st Place, Excellent Research Paper (Science and Technology), Dec 2025, for a paper on design variables and flight stability of an F-22-inspired RC aircraft. 1st Place, School Science Fair (High), Apr 2026.
- Performance Excellence Award, COSMOS (Jul 2026) and Point of View club (Jul 2025). Subject Excellence Award: Precalculus (Dec 2025, Jul 2026) and Creative Writing (Dec 2025).
- 1st Place, English Speaking Contest (Nov 2024) and Social Studies Project Creativity (Jun 2025). 2nd Place, Korean Literature Writing Contest, prose (May 2025). Seven Academic Improvement Awards (2024 to 2026).

RESEARCH AND PROJECTS

F-22-Inspired RC Aircraft: Design and Flight Stability Sep 2025 to Jan 2026

- Designed the airframe in FreeCAD, built and flight-tested a foam-board aircraft, and wrote a 22-page paper relating design variables to stability.

Mechanism of Autopilot (school physics paper)

- Analyzed a control-systems paper and an Airbus patent on autopilot disengagement; compared illustrative P, PI and PID error curves. Literature-based; no flight or hardware testing.

Tern, flight-search web app (flytern.site) Mar 2026 to present

- Built with Next.js, the Duffel API and React Three Fiber in four languages; compares fares, layovers and airport weather. Checkout runs through an external partner.

Automated Drone-Tracking Turret (school science fair)

- Built a camera-tracking rig (Python, OpenCV, Arduino servos); tested YOLOv5 detection and a color-tracking dual-turret demo.

EXPERIENCE

JSL Logistics, *Software Intern* Aug 2026 to present

- Develop staff tools and the public website, including a document-based support chatbot, photo-assisted damage review, and a 3D cargo-loading planner that checks dimensions, stacking and payload limits.

Mintech, Asan, South Korea, *Intern* Aug to Sep 2026

- Assisted with dimensional checks (vernier calipers, micrometers); learned the process flow of sputtering thin-film equipment; completed 4M, 5S and safety onboarding.

Korea Aerospace University Aerospace and Space Camp, *Goyang, Korea* Aug 2026

LEADERSHIP AND ACTIVITIES

COSMOS Aero-Engineering Club, *Co-founder; President since Mar 2026* Sep 2025 to present

- Co-led a rocket thrust-stand project through two static-fire tests; after the first test failed to save data, corrected the recording sequence and captured a continuous thrust record reaching about 190 N.
- Direct Project AEROVIEW, a suction wind tunnel designed in FreeCAD and OpenSCAD, and coordinate TVC-Hopper, which combines reinforcement-learning simulation, avionics and CAD (physical validation in progress).

DAEMUN III, *Deputy Head of Technology* 2026

- Co-developed the conference platform with Junwon Kim (public site, AI assistant, organizer dashboard, delegate submissions with version history) and built the assistant's first backend and admin tools.
- After the assistant stopped during a school-wide demonstration, called a team meeting; the team moved to a gateway-based design with a fallback provider, and the next 51 logged queries recorded no upstream errors.

Student Council, *Planning Dept. Vice President* 2025 to Mar 2026

- Helped plan a joint sports day and ran the live scoreboard at the orientation event.

Mechanics Club, *Vice President* Mar 2026 to present

- Designed and tested a 4-bit ripple-carry adder from logic-gate ICs.

Also: DAEMUN II delegate for Slovakia (2025); Support Group leader (Grade 11); PSYCHOFISH member (2024 to 2025).

SERVICE

- 45 school-certified hours (Nov 2024 to Jun 2026): student council event support, orientation assistance, and graduation ceremony support, including producing the ceremony video. Volunteer, Qingdao Tonghang House (Oct to Nov 2025).

SKILLS

Python, TypeScript, Next.js, React, PostgreSQL, OpenCV, Arduino, FreeCAD. Languages: Korean, English, Mandarin.